We will find an approximation p_n with error less than 10^{-2}.

n	a_n	b_n	p_n	$f(p_n)$	$\frac{1}{2}(b_n - a_n)$
1	0	1	.5	−1.176	.5
2	.5	1	.75	−.412	.25
3	.75	1	.875	+.099	.125
4	.75	.875	.8125	−.169	.0625
5	.8125	.875	.84375	−.0382	.03125
6	.84375	.875	.859375	+.0296	.015625
7	.84375	.859375	.8515625	—	.0078125

We have stopped at $n = 7$, obtaining $c \approx p_7 = .8515625$ with error less than .0078125. This is the first step in which the error is less than 10^{-2}. The decimal place values of p_7 past the second place cannot be taken seriously, but we can conclude that $.843 < c < .860$. □

Bolzano's Theorem

The next result is a generalization of the Location of Roots Theorem. It assures us that a continuous function on an interval takes on (at least once) any number that lies between two of its values.

5.3.7 Bolzano's Intermediate Value Theorem *Let I be an interval and let $f : I \to \mathbb{R}$ be continuous on I. If $a, b \in I$ and if $k \in \mathbb{R}$ satisfies $f(a) < k < f(b)$, then there exists a point $c \in I$ between a and b such that $f(c) = k$.*

Proof. Suppose that $a < b$ and let $g(x) := f(x) - k$; then $g(a) < 0 < g(b)$. By the Location of Roots Theorem 5.3.5 there exists a point c with $a < c < b$ such that $0 = g(c) = f(c) - k$. Therefore $f(c) = k$.

If $b < a$, let $h(x) := k - f(x)$ so that $h(b) < 0 < h(a)$. Therefore there exists a point c with $b < c < a$ such that $0 = h(c) = k - f(c)$, whence $f(c) = k$. Q.E.D.

5.3.8 Corollary *Let $I = [a, b]$ be a closed, bounded interval and let $f : I \to \mathbb{R}$ be continuous on I. If $k \in \mathbb{R}$ is any number satisfying*

$$\inf f(I) \leq k \leq \sup f(I),$$

then there exists a number $c \in I$ such that $f(c) = k$.

Proof. It follows from the Maximum-Minimum Theorem 5.3.4 that there are points c_* and c^* in I such that

$$\inf f(I) = f(c_*) \leq k \leq f(c^*) = \sup f(I).$$

The conclusion now follows from Bolzano's Theorem 5.3.7. Q.E.D.

The next theorem summarizes the main results of this section. It states that the image of a closed bounded interval under a continuous function is also a closed bounded interval. The endpoints of the **image interval** are the absolute minimum and absolute maximum values of the function, and the statement that all values between the absolute minimum and the absolute maximum values belong to the image is a way of describing Bolzano's Intermediate Value Theorem.

5.3.9 Theorem *Let I be a closed bounded interval and let $f: I \to \mathbb{R}$ be continuous on I. Then the set $f(I) := \{f(x) : x \in I\}$ is a closed bounded interval.*

Proof. If we let $m := \inf f(I)$ and $M := \sup f(I)$, then we know from the Maximum-Minimum Theorem 5.3.4 that m and M belong to $f(I)$. Moreover, we have $f(I) \subseteq [m, M]$. If k is any element of $[m, M]$, then it follows from the preceding corollary that there exists a point $c \in I$ such that $k = f(c)$. Hence, $k \in f(I)$ and we conclude that $[m, M] \subseteq f(I)$. Therefore, $f(I)$ is the interval $[m, M]$. Q.E.D.

Warning If $I := [a, b]$ is an interval and $f : I \to \mathbb{R}$ is continuous on I, we have proved that $f(I)$ is the interval $[m, M]$. We have *not* proved (and it is not always true) that $f(I)$ is the interval $[f(a), f(b)]$. (See Figure 5.3.3.) □

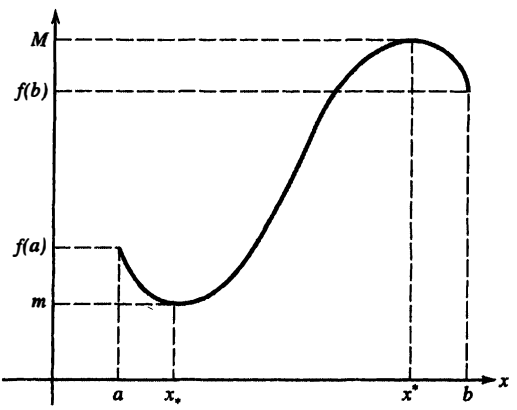


Figure 5.3.3 $f(I) = [m, M]$

The preceding theorem is a "preservation" theorem in the sense that it states that the continuous image of a closed bounded interval is a set of the same type. The next theorem extends this result to general intervals. However, it should be noted that although the continuous image of an interval is shown to be an interval, it is *not* true that the image interval necessarily has the *same form* as the domain interval. For example, the continuous image of an open interval need not be an open interval, and the continuous image of an unbounded closed interval need not be a closed interval. Indeed, if $f(x) := 1/(x^2 + 1)$ for $x \in \mathbb{R}$, then f is continuous on $\mathbb{R}$ [see Example 5.2.3(b)]. It is easy to see that if $I_1 := (-1, 1)$, then $f(I_1) = \left(\frac{1}{2}, 1\right]$, which is not an open interval. Also, if $I_2 := [0, \infty)$, then $f(I_2) = (0, 1]$, which is not a closed interval. (See Figure 5.3.4.)

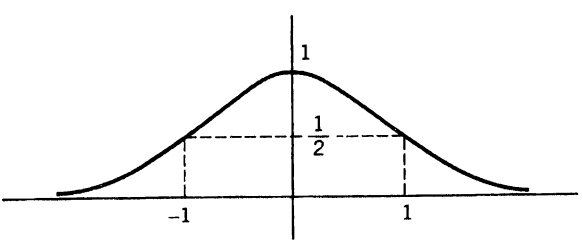


Figure 5.3.4 Graph of $f(x) = 1/(x^2 + 1)$ $(x \in \mathbb{R})$

To prove the Preservation of Intervals Theorem 5.3.10, we will use Theorem 2.5.1 characterizing intervals.

5.3.10 Preservation of Intervals Theorem *Let I be an interval and let $f : I \to \mathbb{R}$ be continuous on I. Then the set $f(I)$ is an interval.*

Proof. Let $\alpha, \beta \in f(I)$ with $a < \beta$; then there exist points $a, b \in I$ such that $\alpha = f(a)$ and $\beta = f(b)$. Further, it follows from Bolzano's Intermediate Value Theorem 5.3.7 that if $k \in (\alpha, \beta)$ then there exists a number $c \in I$ with $k = f(c) \in f(I)$. Therefore $[\alpha, \beta] \subseteq f(I)$, showing that $f(I)$ possesses property (1) of Theorem 2.5.1. Therefore $f(I)$ is an interval. Q.E.D.

Exercises for Section 5.3

1. Let $I := [a, b]$ and let $f : I \to \mathbb{R}$ be a continuous function such that $f(x) > 0$ for each x in I. Prove that there exists a number $\alpha > 0$ such that $f(x) \geq \alpha$ for all $x \in I$.
2. Let $I := [a, b]$ and let $f : I \to \mathbb{R}$ and $g : I \to \mathbb{R}$ be continuous functions on I. Show that the set $E := \{x \in I : f(x) = g(x)\}$ has the property that if $(x_n) \subseteq E$ and $x_n \to x_0$, then $x_0 \in E$.
3. Let $I := [a, b]$ and let $f : I \to \mathbb{R}$ be a continuous function on I such that for each x in I there exists y in I such that $|f(y)| \leq \frac{1}{2}|f(x)|$. Prove there exists a point c in I such that $f(c) = 0$.
4. Show that every polynomial of odd degree with real coefficients has at least one real root.
5. Show that the polynomial $p(x) := x^4 + 7x^3 - 9$ has at least two real roots. Use a calculator to locate these roots to within two decimal places.
6. Let f be continuous on the interval $[0, 1]$ to $\mathbb{R}$ and such that $f(0) = f(1)$. Prove that there exists a point c in $[0, \frac{1}{2}]$ such that $f(c) = f\left(c + \frac{1}{2}\right)$. [*Hint*: Consider $g(x) = f(x) - f\left(x + \frac{1}{2}\right)$.] Conclude that there are, at any time, antipodal points on the earth's equator that have the same temperature.
7. Show that the equation $x = \cos x$ has a solution in the interval $[0, \pi/2]$. Use the Bisection Method and a calculator to find an approximate solution of this equation, with error less than 10^{-3}.
8. Show that the function $f(x) := 2 \ln x + \sqrt{x} - 2$ has root in the interval $[1, 2]$, Use the Bisection Method and a calculator to find the root with error less than 10^{-2}.
9. (a) The function $f(x) := (x-1)(x-2)(x-3)(x-4)(x-5)$ has five roots in the interval $[0, 7]$. If the Bisection Method is applied on this interval, which of the roots is located?
   (b) Same question for $g(x) := (x-2)(x-3)(x-4)(x-5)(x-6)$ on the interval $[0, 7]$.
10. If the Bisection Method is used on an interval of length 1 to find p_n with error $|p_n - c| < 10^{-5}$, determine the least value of n that will assure this accuracy.
11. Let $I := [a, b]$, let $f : I \to \mathbb{R}$ be continuous on I, and assume that $f(a) < 0, f(b) > 0$. Let $W := \{x \in I : f(x) < 0\}$, and let $w := \sup W$. Prove that $f(w) = 0$. (This provides an alternative proof of Theorem 5.3.5.)
12. Let $I := [0, \pi/2]$ and let $f : I \to \mathbb{R}$ be defined by $f(x) := \sup\{x^2, \cos x\}$ for $x \in I$. Show there exists an absolute minimum point $x_0 \in I$ for f on I. Show that x_0 is a solution to the equation $\cos x = x^2$.
13. Suppose that $f : \mathbb{R} \to \mathbb{R}$ is continuous on $\mathbb{R}$ and that $\lim_{x \to -\infty} f = 0$ and $\lim_{x \to \infty} f = 0$. Prove that f is bounded on $\mathbb{R}$ and attains either a maximum or minimum on $\mathbb{R}$. Give an example to show that both a maximum and a minimum need not be attained.
14. Let $f : \mathbb{R} \to \mathbb{R}$ be continuous on $\mathbb{R}$ and let $\beta \in \mathbb{R}$. Show that if $x_0 \in \mathbb{R}$ is such that $f(x_0) < \beta$, then there exists a δ-neighborhood U of x_0 such that $f(x) < \beta$ for all $x \in U$.